

Energy-Efficient Active-RIS-Assisted RSMA-ISAC Under Imperfect CSI and Realistic Propagation Environments

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1 Problem Statement

Future wireless networks are expected to support both high-rate communication and accurate sensing using the same wireless infrastructure. This concept is known as Integrated Sensing and Communication (ISAC).

Reconfigurable Intelligent Surfaces (RISs) can improve wireless propagation by controlling the wireless environment. An active RIS further incorporates amplification, allowing the surface to compensate for signal attenuation. For multi-user communication, Rate-Splitting Multiple Access (RSMA) provides flexible interference management by dividing information into common and private streams.

The foundation paper for this project, “Beamforming Design for Enhanced Sum Rate in Active RIS-RSMA-Assisted ISAC Systems Under Imperfect CSIT/CSIR,” combines active RIS, RSMA, ISAC, and imperfect CSI and develops a joint beamforming and precoding optimization framework. Its primary objective is to improve communication sum rate.

However, practical wireless environments are more complex than controlled mathematical simulation environments. Propagation can be affected by user and target distances, LOS/NLOS conditions, obstacles, RIS location, path loss, fading, and other channel characteristics. Furthermore, active RIS requires additional amplification power and introduces additional noise, making energy-efficient operation an important consideration.

Therefore, this project will retain the foundation system

$$\boxed{\text{RSMA} + \text{Active RIS} + \text{ISAC} + \text{Imperfect CSI}} \quad (1)$$

and extend its evaluation in two directions:

1. energy-efficient operation, and
2. realistic propagation environments.

The main research problem is:

How does the performance and energy efficiency of the active-RIS RSMA-ISAC system change when the controlled channel assumptions are replaced by more realistic propagation conditions, while imperfect CSI is retained?

Assumptions and Scope

The project will initially use realistic geometry and channel simulation rather than promising real terrain or measured-channel data. The specific realistic channel models will be selected according to the available simulator and data.

The study will focus on:

- active-RIS-assisted RSMA-ISAC,
- imperfect CSI,
- energy efficiency,
- user and target distances,
- LOS/NLOS propagation,
- channel characteristics,
- RIS position and possible obstacles, and
- the trade-off between communication performance, sensing performance, and power consumption.

The active-RIS versus passive-RIS comparison will not be treated as the main contribution because this comparison is already investigated in the foundation paper. :contentReference[oaicite:1]index=1

2 Review of Existing Work

The proposed work is based on four closely related research directions.

2.1 Active RIS and RSMA

Niu et al. investigated an active-RIS-assisted RSMA network with a focus on the trade-off between spectral efficiency and energy efficiency. Their work demonstrates that active RIS and RSMA can be studied together while considering energy consumption.

2.2 Active RIS and ISAC

Ye et al. studied energy-efficiency optimization in active-RIS-assisted ISAC systems. This work highlights the importance of considering the power consumed by active RIS together with communication and sensing performance.

2.3 RSMA, Active RIS and Energy Efficiency

Truong et al. investigated energy efficiency in RSMA-enhanced active RIS systems. Their work further establishes energy efficiency as an important consideration when active RIS and RSMA are jointly used.

2.4 Active RIS, RSMA, ISAC and Imperfect CSI

Parihar et al. proposed the foundation system used in this project. The system combines active RIS, RSMA, ISAC, and imperfect CSIT/CSIR. The authors formulate a joint beamforming and precoding optimization problem to enhance the communication sum rate.

The imperfect CSI model can be represented as

$$\hat{\mathbf{H}} = \mathbf{H} + \mathbf{E}, \quad (2)$$

where $\mathbf{H}$ is the actual channel, $\hat{\mathbf{H}}$ is the estimated channel, and $\mathbf{E}$ represents channel estimation error.

The optimization problem is non-convex and is addressed using methods including alternating optimization, semidefinite relaxation, CCCP, and auxiliary-variable techniques.

The foundation paper also evaluates several system baselines, including passive RIS and no-RIS configurations. Therefore, the proposed project focuses on extending the system rather than repeating the same active-versus-passive comparison.

3 Technical Progress and SOTA Positioning

3.1 Technical Progress

At the current stage, the project has established the research foundation, system architecture, research gap, performance metrics, and experimental direction.

The starting system is

$$\boxed{\text{RSMA} + \text{Active RIS} + \text{ISAC} + \text{Imperfect CSI}} \quad (3)$$

The proposed extension is

$$\boxed{\text{Foundation System} + \text{Energy Efficiency} + \text{Realistic Propagation}} \quad (4)$$

The development will be carried out incrementally:

$$\boxed{\text{Foundation Model} \rightarrow \text{Baseline Reproduction} \rightarrow \text{EE Analysis} \rightarrow \text{Realistic Channels} \rightarrow \text{Imperfect CSI}} \quad (5)$$

The first technical objective is to reproduce and verify the foundation model. After establishing the baseline, energy-efficiency analysis and realistic propagation conditions will be introduced.

3.2 Energy-Efficiency Extension

The project will evaluate the trade-off between communication performance and power consumption.

The energy-efficiency metric is defined as

$$\boxed{EE = \frac{R_{\text{sum}}}{P_{\text{total}}}} \quad (6)$$

where R_{sum} is the communication sum rate and P_{total} is the total system power consumption.

The total power consumption is represented as

$$\boxed{P_{\text{total}} = P_{\text{BS}} + P_{\text{RIS}} + P_{\text{circuit}}} \quad (7)$$

This allows the project to investigate whether operating at a lower transmit power or lower active-RIS amplification can provide better energy efficiency while maintaining acceptable communication and sensing performance.

3.3 Realistic Propagation Extension

The realistic propagation study will investigate parameters such as

$$d_{\text{BS-RIS}}, \quad d_{\text{RIS-user}}, \quad d_{\text{RIS-target}}. \quad (8)$$

The effects of these parameters will be evaluated in terms of:

- communication rate,
- sensing performance,
- required RIS amplification,
- power consumption, and
- energy efficiency.

The project will also investigate different propagation conditions, including:

- LOS-dominated channels,
- NLOS channels,
- different path-loss conditions,
- different fading characteristics, and
- different RIS positions.

Where appropriate, obstacles and blockage conditions may also be introduced.

3.4 Imperfect CSI Extension

Imperfect CSI will remain part of the system rather than being removed when realistic propagation conditions are introduced.

The channel estimation model is

$$\boxed{\hat{\mathbf{H}} = \mathbf{H} + \mathbf{E}} \quad (9)$$

The channel estimation error can be controlled using

$$\sigma_e^2. \quad (10)$$

Different error levels will be considered after a stable baseline simulation is established. The main objective is to study the combined effect of

$$\boxed{\text{Realistic Propagation} + \text{Imperfect CSI}} \quad (11)$$

on communication, sensing, and energy efficiency.

3.5 SOTA Positioning

Table 1 summarizes the position of the proposed project relative to the selected related works.

Work	Active RIS	RSMA	ISAC	Imperfect CSI	EE
Niu et al. (2023)	Yes	Yes	No	–	Yes
Ye et al. (2025)	Yes	No	Yes	–	Yes
Truong et al. (2025)	Yes	Yes	No	–	Yes
Parihar et al. (2026)	Yes	Yes	Yes	Yes	No
Proposed Project	Yes	Yes	Yes	Yes	Yes

Table 1: SOTA positioning of the proposed project.

The SOTA comparison shows that the foundation paper provides the closest starting point because it combines active RIS, RSMA, ISAC, and imperfect CSI. The proposed project retains these components while adding a specific focus on energy efficiency and realistic propagation conditions.

The proposed contribution is therefore not the introduction of a new wireless architecture. Instead, it is an extended evaluation of the existing architecture under practical propagation and energy-efficiency considerations.

3.6 Research Gap

The main research gap identified from the reviewed work is the need to evaluate the active-RIS-assisted RSMA-ISAC system beyond the controlled simulation conditions of the foundation work.

The proposed project addresses this gap through:

1. energy-efficiency analysis,
2. realistic propagation conditions,
3. continued consideration of imperfect CSI, and
4. analysis of the interaction between these factors.

The project will compare the controlled foundation simulation with realistic propagation scenarios to determine whether the performance trends predicted by the mathematical model remain valid in more practical conditions.

4 Research Questions

The project addresses the following three research questions.

RQ1 – Energy Efficiency

How can the energy efficiency of the active-RIS-assisted RSMA-ISAC system be improved while maintaining acceptable communication and sensing performance?

RQ2 – Realistic Propagation

How do user/target distance, LOS/NLOS propagation, and channel conditions affect the performance of the system compared with the controlled simulation environment of the foundation paper?

RQ3 – Imperfect CSI

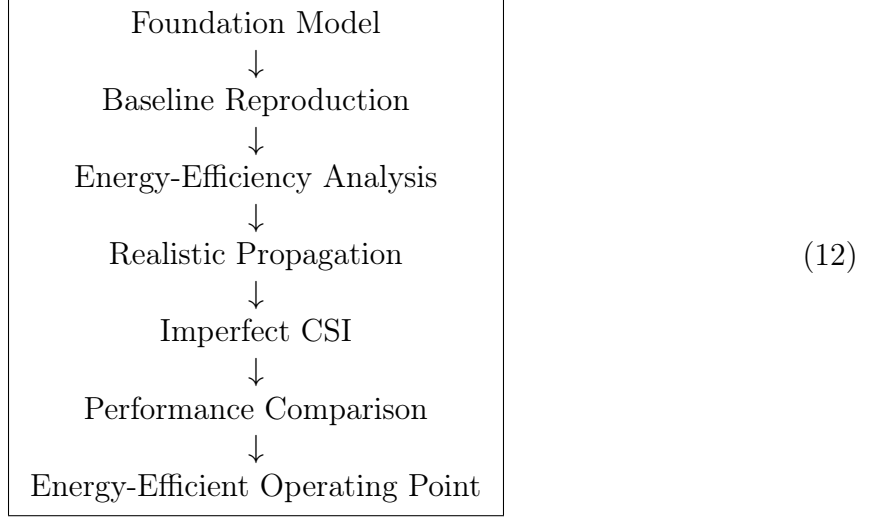
How does CSI estimation error affect the communication, sensing, and energy-efficiency performance of the system under realistic propagation conditions?

5 Methodology

The project will follow an incremental experimental methodology.

1. **Foundation model:** Implement the active-RIS-assisted RSMA-ISAC framework described in the foundation paper.
2. **Baseline verification:** Reproduce the baseline behavior and compare the implementation with available results from the foundation paper.
3. **Energy-efficiency analysis:** Add total power consumption and evaluate energy efficiency for different transmit-power and active-RIS amplification settings.
4. **Realistic propagation:** Introduce realistic distances, LOS/NLOS conditions, channel characteristics, RIS positions, and possible obstacles according to the available simulation environment.
5. **Imperfect CSI:** Vary the channel-estimation error and evaluate its effect under realistic propagation conditions.
6. **Performance comparison:** Compare the controlled foundation environment with realistic propagation conditions.
7. **Analysis:** Identify operating conditions that provide a suitable balance between communication rate, sensing performance, power consumption, and energy efficiency.

The overall methodology can be summarized as



5.1 Performance Metrics

The main performance metrics are:

Communication Rate

$$R_{\text{sum}} = \sum_k R_k \quad (13)$$

where

$$R_k = B \log_2(1 + \text{SINR}_k). \quad (14)$$

Sensing Performance

The initial sensing metric is

$$\text{SNR}_{\text{sense}} = \frac{P_{\text{signal}}}{P_{\text{noise}}} \quad (15)$$

The final sensing metric will depend on the sensing model used in the implementation.

Power Consumption

$$P_{\text{total}} = P_{\text{BS}} + P_{\text{RIS}} + P_{\text{circuit}} \quad (16)$$

Energy Efficiency

$$EE = \frac{R_{\text{sum}}}{P_{\text{total}}} \quad (17)$$

5.2 Planned Comparisons

The primary comparisons in the project are:

1. **Controlled vs. realistic environment**

$$\text{Foundation Simulation} \quad \text{vs.} \quad \text{Realistic Propagation} \quad (18)$$

2. **CSI quality**

$$\text{Low CSI Error} \quad \text{vs.} \quad \text{High CSI Error} \quad (19)$$

3. **Propagation conditions**

$$\text{LOS} \quad \leftrightarrow \quad \text{NLOS} \quad (20)$$

together with different distances and channel conditions.

4. **Energy-efficiency operating points**

Different BS transmit powers and active-RIS amplification levels will be compared to identify suitable rate–power trade-offs.

6 Expected Contribution

The expected contribution of the project is:

An extended evaluation of an active-RIS-assisted RSMA-ISAC system under imperfect CSI using more realistic propagation conditions, together with an analysis of energy-efficient operating conditions.

The project is expected to:

- establish the foundation active-RIS-assisted RSMA-ISAC model,
- evaluate its energy-efficiency behavior,
- investigate realistic propagation effects,
- study the interaction between realistic propagation and imperfect CSI, and
- identify operating conditions that provide a suitable balance between communication performance, sensing performance, and energy consumption.

7 Timeline

Stage	Planned Work
M1	Problem definition, literature review, research gap identification, and SOTA positioning
M2	Implementation and reproduction of the foundation active-RIS RSMA-ISAC model
M3	Energy-efficiency model and introduction of realistic propagation/channel conditions
M4	Simulation, parameter analysis, and imperfect CSI evaluation
M5	Result analysis, final optimization/improvement, report preparation, and presentation

Table 2: Planned project timeline.

8 References

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